Algebra Essentials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

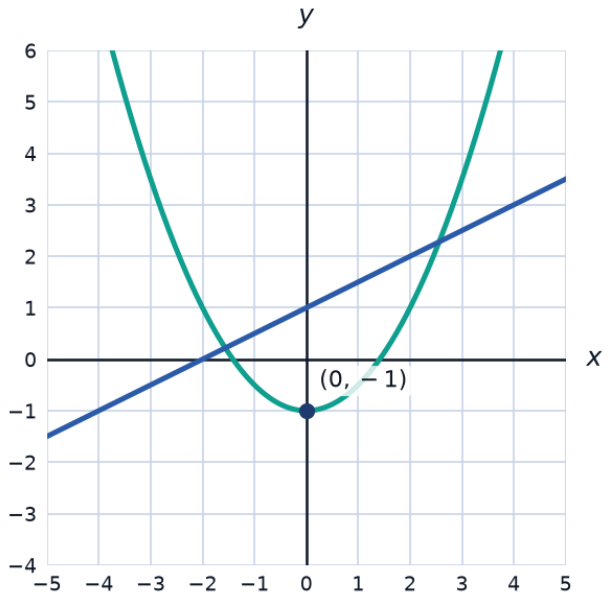
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating expressions

If $x = 4$, evaluate $3x + 7$.

2 [Foundational] • Target: Evaluating expressions

If $a = 5$ and $b = 2$, evaluate $a^2 - 3b$.



3 [Foundational] • Target: Evaluating expressions

If $x = -3$, evaluate $2x^2 + 5x$.

4 [Foundational] • Target: Evaluating expressions

If $p = 6$ and $q = -2$, evaluate $\frac{p - q}{q + 6}$.

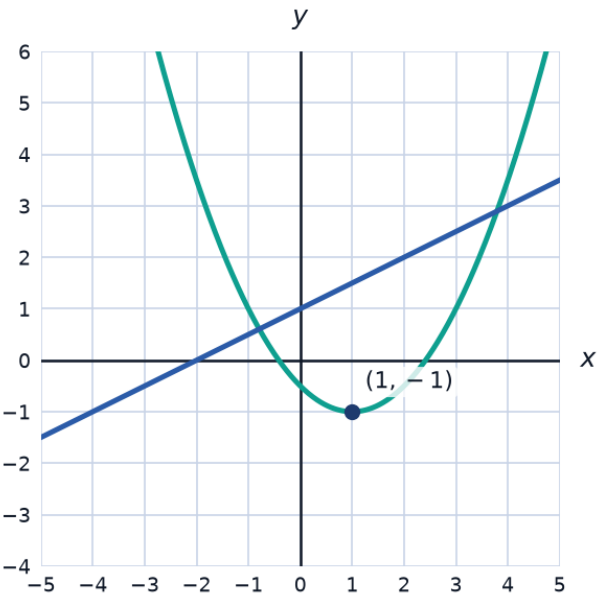
5 [Foundational] • Target: Solving linear equations

Solve $2x + 5 = 17$.

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6 [Foundational] • Target: Solving linear equations

Solve $5x - 8 = 27$.



7 [Foundational] • Target: Solving linear equations

Solve $3(x - 4) = 21$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving linear equations

Solve $\frac{x}{4} + 3 = 10$.

9 [Intermediate] • Target: Solving linear equations

Solve $7x + 3 = 4x + 18$.

10 [Intermediate] • Target: Expanding and simplifying

Expand $(x + 3)(x + 5)$.

11 [Intermediate] • Target: Expanding and simplifying

Expand $(x - 4)(x + 6)$.

12 [Intermediate] • Target: Expanding and simplifying

Expand $(2x + 1)(x - 3)$.

13 [Intermediate] • Target: Expanding and simplifying

Simplify $3(2x - 5) - 2(x - 4)$.

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14 [Intermediate] • Target: Factoring

Factor $x^2 + 9x + 20$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Factoring • Cross-Lesson: 7.01 Data Analysis & Probability

Factor $x^2 - 2x - 15$.

16 [Challenging] • Target: Factoring • Cross-Lesson: 8.01 Word Problems & Reasoning

Factor $x^2 - 49$.

17 [Challenging] • Target: Factoring • Cross-Lesson: 1.01 Arithmetic & Number Sense

Factor $6x + 15$.

18 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Solve $x^2 + 5x + 6 = 0$.

19 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 4.01 Geometry & Measurement

Solve $x^2 - 7x + 12 = 0$.

20 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 5.01 Quantitative Comparisons

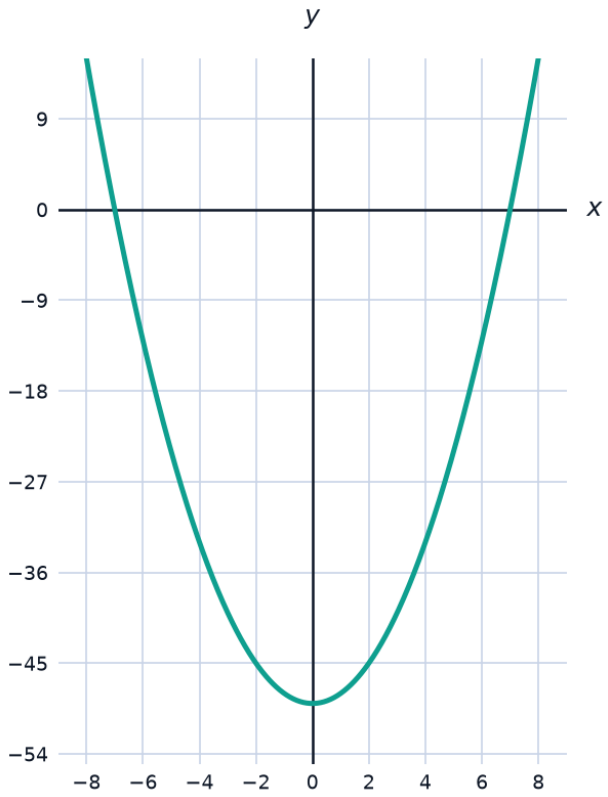
Solve $x^2 - 16 = 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: MCQ — Solving quadratic equations

The graph shows $y = x^2 - 49$. What are its roots (where the curve crosses the x-axis)?



a. x = 7 only

b. x = 49

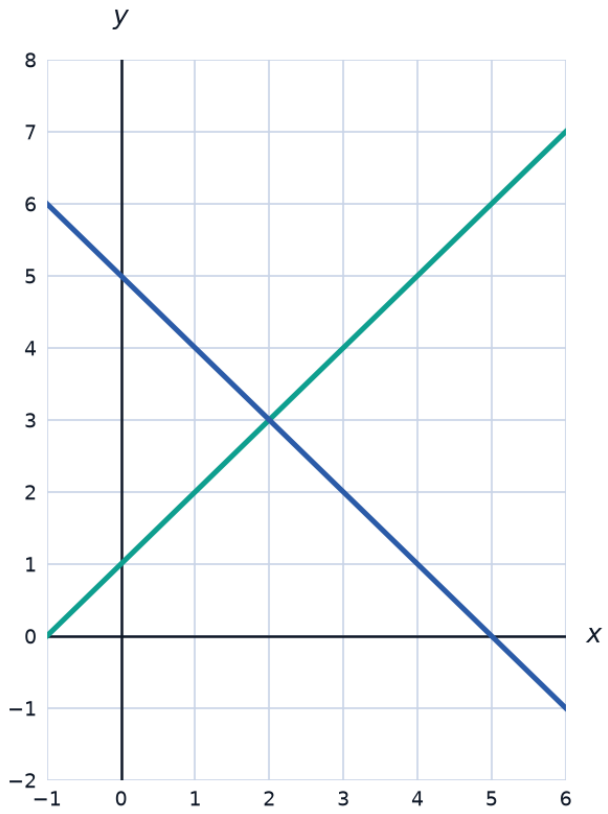
c. x = 7 or x = -7

d. x = -7 only

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22 [Foundational] • Target: MCQ — Systems of two equations

The graph shows two lines. What is their point of intersection?



a. (1, 4)

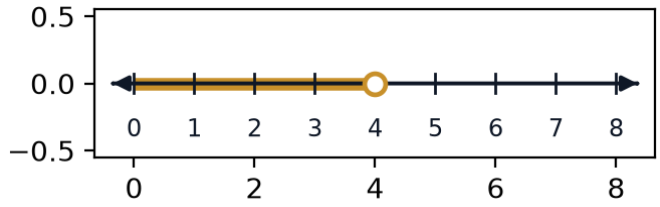
b. (0, 5)

c. (2, 3)

d. (3, 2)

23 [Foundational] • Target: MCQ — Inequalities

The number line shows the solution set of an inequality. Which inequality matches?



a. $x \geq 4$

b. $x \leq 4$

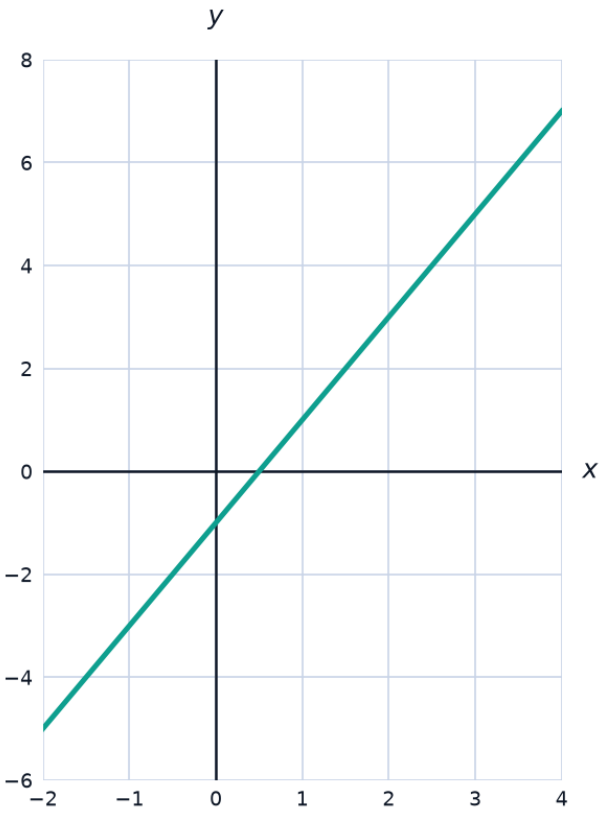
c. $x > 4$

d. $x < 4$

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24 [Foundational] • Target: MCQ — Reading function graphs

The graph shows a line. What is its slope?



a. $\frac{1}{2}$

b. 2

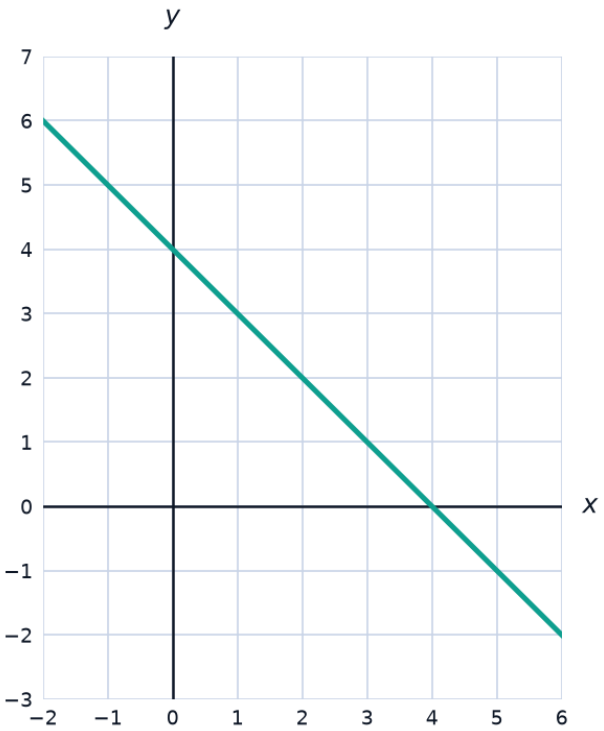
c. -2

d. -1

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25 [Foundational] • Target: MCQ — Reading function graphs

What is the y-intercept of the line shown?



a. 4

b. 0

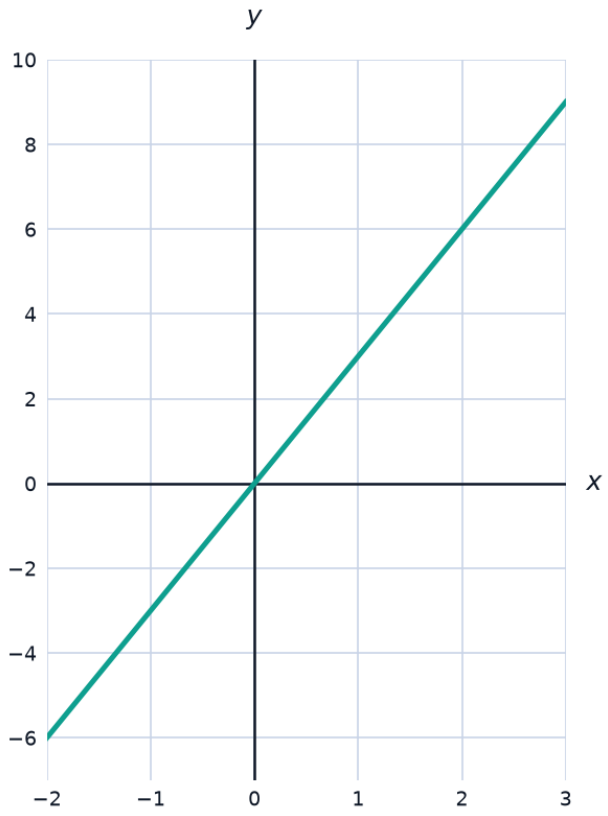
c. −4

d. 1

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26 [Foundational] • Target: MCQ — Reading function graphs

What is the equation of the line shown?



a. $y = 3x + 1$

b. $y = x + 3$

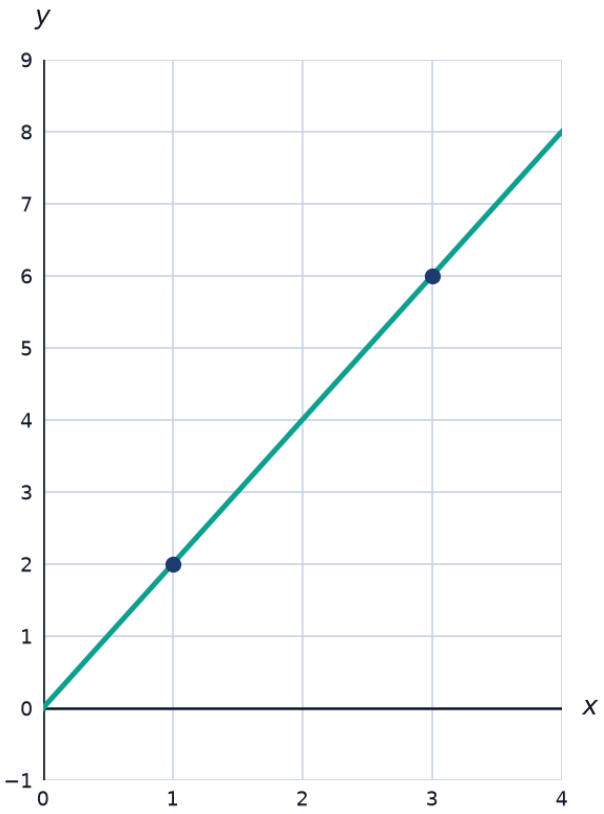
c. $y = \frac{1}{3}x$

d. $y = 3x$

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27 [Foundational] • Target: MCQ — Reading function graphs

Find the slope of the line through the two marked points.



a. 2

b. $\frac{1}{2}$

c. -2

d. 4

28 [Foundational] • Target: MCQ — Substitution into formulas

The rectangle shown has area $A = lw = 48$ and length $l = 8$. Find the width w.

a. 8

b. 6

c. 384

d. 40

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: MCQ — Evaluating expressions

If $x = 6$, evaluate $4x - 9$.

a. 33

b. 24

c. -15

d. 15

30 [Intermediate] • Target: MCQ — Solving linear equations

Solve $4(x + 2) = 3x + 15$.

a. $x = 1$

b. $x = 23$

c. $x = -7$

d. $x = 7$

31 [Intermediate] • Target: MCQ — Expanding and simplifying

Expand $(x - 5)^2$.

a. $x^2 + 10x + 25$

b. $x^2 - 5x + 25$

c. $x^2 - 10x + 25$

d. $x^2 - 25$

32 [Intermediate] • Target: MCQ — Factoring

Factor $x^2 + 3x - 18$.

a. $(x + 2)(x - 9)$

b. $(x - 6)(x + 3)$

c. $(x + 9)(x - 2)$

d. $(x + 6)(x - 3)$

33 [Intermediate] • Target: MCQ — Systems of two equations

Solve the system: $x + 2y = 8$, $x - y = 2$.

a. x = 2, y = 3

b. x = 6, y = 1

c. x = 4, y = 2

d. x = 8, y = 0

34 [Intermediate] • Target: MCQ — Laws of exponents

Simplify $x^0 \cdot x^4$ $(x \neq 0)$.

a. 1

b. x^5

c. x^0

d. x^4

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35 [Intermediate] • Target: MCQ — Substitution into formulas

Given $F = ma$, find F when $m = 7$ and $a = 5$.

a. 35

b. 75

c. 12

d. 2.4

36 [Intermediate] • Target: MCQ — Word problems

A number increased by 20% equals 72. Find the number.

a. 90

b. 52

c. 60

d. 86.4

37 [Intermediate] • Target: MCQ — Systems of two equations

Solve $9x + 12 = -24$.

a. -4

b. -2

c. 4

d. -6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Laws of exponents • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Solve $9x + 1 = 28$.

a. -3

b. 3

c. 5

d. 1

39 [Challenging] • Target: MCQ — Inequalities • Cross-Lesson: 4.01 Geometry & Measurement

Solve $5x + 18 = -27$.

a. -9

b. 9

c. -7

d. -11

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40 [Challenging] • Target: MCQ — Reading function graphs • Cross-Lesson: 5.01 Quantitative Comparisons

Expand $(x+3)(x+7)$.

a. $x^2+21x+10$

b. $x^2-10x+21$

c. $x^2+10x-21$

d. $x^2+10x+21$

41 [Challenging] • Target: MCQ — Substitution into formulas • Cross-Lesson: 6.01 Sequences & Patterns

Expand $(x-2)(x-1)$.

a. x^2+2x-3

b. x^2+3x+2

c. x^2-3x+2

d. x^2-3x-2

42 [Challenging] • Target: MCQ — Translating words into equations • Cross-Lesson: 7.01 Data Analysis & Probability

Expand $(x+8)(x-8)$.

a. $x^2+2x-64$

b. x^2-64

c. x^2-64x

d. x^2+64

43 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 8.01 Word Problems & Reasoning

Solve $x^2-25=0$.

a. $5,\ -5$

b. $-5, 5$

c. $6,\ -5$

d. $5,\ -4$

44 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Arithmetic & Number Sense

Solve the system: $x+y=-3$, $x-y=9$.

a. $3,\ -4$

b. $-6, 3$

c. $3,\ -6$

d. $5,\ -6$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Simplify $(b^4)^3$.

a. b^{12}

b. b^7

c. b^{14}

d. b^{10}

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